

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2024****Course: B.Tech Branch : Common to all Branches Semester : II****Subject Code & Name: BTES205E (Energy and Environment Engineering)****Max Marks: 60****Date: 30/12/2024****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/ CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	Steam Power Plant is also known as -----.	CO1	1
	a. Thermal Power Plant b. Fossil Fuels c. Boilers d. Turbines		
2	A Steam power plant converts the Chemical Energy of the fossil fuels into -----.	CO2	1
	a. Chemical Energy b. Mechanical/Electrical Energy c. Steam d. All of these		
3	A Steam Turbine is used to convert the heat contained in steam into -----.	CO3	1
	a. Nuclear Chain Reaction b. Thermal Power Station c. Electricity d. Mechanical Energy		
4	In a Gas Turbine Power Plant, ----- is used as the working agent.	CO2	1
	a. Electrical Energy b. Compressor c. Combustion Chamber d. Natural Air		
5	A Thermoelectric Generator is an Electric device that, converts Heat Energy produced from a Heat source directly into ----- .	CO1	1

	a. Electrical Energy	b. Heat	c. Steam	d. Electricity		
6	Ocean Thermal Energy Conversion is an ----- Generation System.				CO3	1
	a. Biogas	b. Biomass	c. Electricity	d. Heat		
7	The Energy obtained from the rise and fall of tides is called -----				CO1	1
	a. Ocean Thermal Energy	b. Clean Energy	c. Tidal Energy	d. Fossil Fuels		
8	A Fuel Cell is an Electrochemical Cell that, generates Electrical Energy from Fuel via an ----- .				CO2	1
	a. Electrochemical Reaction	b. High Efficiency	c. Sodium Hydroxide	d. Carbon Electrodes		
9	Fuel Cells, However, offer a platform for the direct conversion of Chemical Energy into -----.				CO3	1
	a. Steam	b. Electric Current	c. Electricity	d. Electrical Energy		
10	The MHD Generation, also known as Magneto Hydrodynamic power generation, is a direct Energy Conversion system which, converts the Heat Energy into -----.				CO4	1
	a. Fuel Cell	b. Steam	c. Electricity	d. Electrical Energy		
11	----- is a unique substance, because, of it's property of self renew and Cleaning.				CO5	1
	a. Air	b. Water	c. Fuel	d. All of these		
12	Water pollution is the second most imperative Environmental concern along with ----- .				CO5	1
	a. Air Pollution	b. Soil Pollution	c. Thermal Pollution	d. All of these		

Q. 2	Solve the following.		12
A)	Explain the working of Simple Gas Turbine power plant using neat diagram.	CO1	6
B)	Write the working of a Hydro Electric Power Plant with neat diagram.	CO2	6
Q.3	Solve the following.		12
A)	Write the disadvantages and advantages of using energy from water.	CO2	6
B)	With neat diagram, Explain the working of Magneto Hydro Dynamic Generator.	CO3	6
Q. 4	Solve Any Two of the following.		12
A)	Explain Energy Conservation. Report five ways we could conserve Energy individually or collectively.	CO3	6
B)	Write the benefits of Energy Efficiency.	CO2	6
C)	Explain the Energy Pyramid with neat diagram.	CO1	6
Q.5	Solve Any Two of the following.		12
A)	Short notes:- i) Primary pollutants ii) Secondary Pollutants	CO1	6
B)	Write the Effect of Air pollution on plants.	CO2	6
C)	Write in detail the Effect of Air pollution on climate.	CO3	6
Q. 6	Solve Any Two of the following.		12
A)	Write in detail the causes of water pollution.	CO3	6
B)	Write the sources of Noise pollution in detail.	CO4	6
C)	Write the control measures of Soil pollution.	CO5	6
*** End ***			